HW #

1. (a) $y(n) = \log[x(n)]$

① $\log[\alpha x_1(n) + \beta x_2(n)] \neq \alpha \log[x_1(n)] + \beta \log[x_2(n)]$, so it is **nonlinear**

② $y(n-k) = \log[x(n-k)]$, so it is **time-invariant**

③ **Causal** since the value of $y(n_0)$ just depends on the value of input $n = n_0$

(b) $y(n) = x(n)\,x(n-2)$

① $\alpha x_1(n)x_1(n-2) + \beta x_2(n)x_2(n-2) \neq [\alpha x_1(n) + \beta x_2(n)][\alpha' x_1(n-2) + \beta x_2(n-2)]$, so it is **nonlinear**

② $y(n-k) = x(n-k)x(n-k-2)$, so it is **time invariant.**

③ **causal**, since, the $y(n)$ only depends on the input at n_0, and n_0-2.

(c) $y(n) = \sum_{k=0}^{n} x(k)$

①. $\alpha \sum_{k=0}^{n} x_1(k) + \beta \sum_{k=0}^{n} x_2(k) = \sum_{k=0}^{n} [\alpha x_1(k) + \beta x_2(k)]$, so it is **linear.**

② $\sum_{k=0}^{n} x(k-m)$ let $k' = k-m$, $\Rightarrow \sum_{k'=-m}^{n-m} x(k') \neq y(n-m) = \sum_{k=0}^{n-m} x(k)$, so it is **time-varying.**

③ **Causal** since, only depends on $n \leq n_0$ of $x(n)$.

(d). $y(n) = \sum_{k=-\infty}^{n} x(k)$

① $\alpha \sum_{k=-\infty}^{n} x_1(k) + \beta \sum_{k=-\infty}^{n} x_2(k) = \sum_{k=-\infty}^{n} [\alpha x_1(k) + \beta x_2(k)]$, so it is **linear.**

② $\sum_{k=-\infty}^{n} x(k-m)$ let $k' = k-m$, $\Rightarrow \sum_{k=-\infty}^{n-m} x(k') = y(n-m) = \sum_{k=-\infty}^{n-m} x(k)$, so it is **time-invariant.**

③ **Causal.** $y(n_0)$ depends on $n \leq n_0$ of $x(n)$.

2. (a) $h(n) = \left(\frac{3}{4}\right)^n u(n-1)$

① since $h(n) = 0$ for $n < 0$, so it is **Causal.**

② $\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=1}^{\infty} \left|\left(\frac{3}{4}\right)^n\right| = \frac{\frac{3}{4}}{1-\frac{3}{4}} = 3 < \infty$, so it is **BIBO stable.**

(b) $h(n) = (2)^n u(-n+1)$

① since $h(n) \neq 0$, for $n < 0$. So it is **non-causal.**

② $\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=-\infty}^{1} 2^n \overset{m=-n}{=\!=} \sum_{m=-1}^{\infty} \left(\frac{1}{2}\right)^m = \frac{\left(\frac{1}{2}\right)^{-1}}{1-\frac{1}{2}} = 4 < \infty$, so it is **BIBO stable**

3. $h(n) = [h_1(n) \otimes h_2(n) - h_3(n)] \otimes h_4(n)$

Since $h_1(n) = (\frac{1}{2})^n u(n)$, $h_2(n) = \delta(n)$, $h_1(n) \otimes h_2(n) = (\frac{1}{2})^n u(n) \otimes \delta(n) = (\frac{1}{2})^n u(n)$

$h(n) = [(\frac{1}{2})^n u(n) - (\frac{1}{3})^n u(n)] \otimes (\frac{1}{3})^n u(n)$

$= \underbrace{(\frac{1}{2})^n u(n) \otimes (\frac{1}{3})^n u(n)}_{\alpha} - \underbrace{(\frac{1}{3})^n u(n) \otimes (\frac{1}{3})^n u(n)}_{\beta}$

$\alpha = \sum_{k=-\infty}^{\infty} x_1(k) h(n-k) = \sum_{k=-\infty}^{\infty} (\frac{1}{2})^k \underbrace{u(k)}_{k \geq 0} \cdot (\frac{1}{3})^{n-k} \cdot \underbrace{u(n-k)}_{\substack{n-k \geq 0 \\ k \leq n}} = \sum_{k=0}^{n} (\frac{1}{2})^k \cdot (\frac{1}{3})^{n-k} = (\frac{1}{3})^n \sum_{k=0}^{n} (\frac{3}{2})^k$

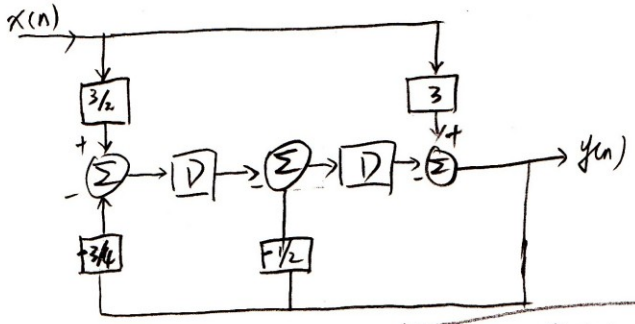
$= (\frac{1}{3})^n \frac{1-(\frac{3}{2})^{n+1}}{1-\frac{3}{2}}$

$\beta = \sum_{k=-\infty}^{\infty} (\frac{1}{3})^k \cdot u(k) \cdot (\frac{1}{3})^{n-k} \cdot u(n-k) = \sum_{k=0}^{n} (\frac{1}{3})^k \cdot (\frac{1}{3})^{n-k} = \sum_{k=0}^{n} (\frac{1}{3})^n = (n+1)(\frac{1}{3})^n = -2(\frac{1}{3})^n + 3 \cdot (\frac{1}{2})^n$

$\Rightarrow h(n) = (-2(\frac{1}{3})^n + 3 \cdot (\frac{1}{2})^n - (n+1)(\frac{1}{3})^n u(n) = 3 \cdot (\frac{1}{2})^n - (n+3)(\frac{1}{3})^n u(n)$

4. (a). $(\frac{1}{3}) y(n) - (\frac{1}{6}) y(n-1) - \frac{1}{6} y(n-2) = x(n) + \frac{1}{2} x(n-2)$

$\Rightarrow y(n) - \frac{1}{2} y(n-1) - \frac{1}{6} y(n-2) = 3x(n) + \frac{3}{2} x(n-2)$



(b). $h(n) = [2, 1, 4, 5]$

$y(n) = h(n) \otimes x(n) = \sum_{k=0}^{3} h(k) x(n-k) = 2x(n) + x(n-1) + 4x(n-2) + 5x(n-3)$

